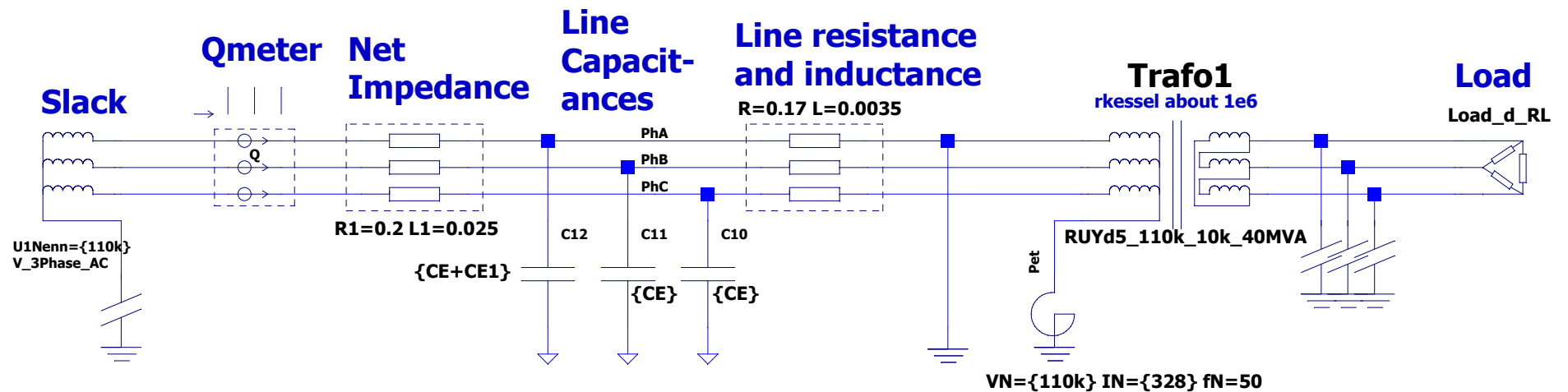


Middle voltage transformer with petersen 328A (Transient analysis) wo fault

```
.param last=20e6
.param last=20e-6
.param tdv=1 tdt=1 ce=5012n ce1=10n fn1=50

.options vntol=1m abstol=1u
.tran 0 5s 0 0.1m
.ac lin 1000 40 52
```



$$XC = 1 / \omega / C = 1 / 314.159 / 5012e-9 = 635.10 \Omega$$

$$I_0 = U_N / 1.73205 / 635.10 \Omega = 100 \text{ A}$$

$$I_E = 3 * I_0 = 300 \text{ A (only for cables)}$$

To see maximal petersen voltage
at resonant point
unset fault (delete wire or set rgnd=1e6) and measure petersen voltage

To see minimal fault current
at resonant point
enable fault and measure fault current